

YEAR 11 YEARLY EXAMINATION
PRELIMINARY HSC ASSESSMENT TASK 3
SEPTEMBER 2009

- Working time allowed – 120 minutes
- Write using black or blue pen
- Approved calculators may be used
- All necessary working should be shown
- Start each question on a new page
- Attempt all questions
- Questions are of equal value
- Full marks may not be awarded if working is poorly set out or difficult to read

Teacher: _____

[illegible]

Question 1 (10 marks)

- a) Evaluate $3^{2.5}$ correct to 3 significant figures 2
- b) Factorise completely $ab - a - bx + x$ 2
- c) Find integers a and b such that $\frac{1}{\sqrt{3}+2} = a\sqrt{3} + b$ 2
- d) Solve $\tan\theta = \frac{-1}{\sqrt{3}}$ for $0^\circ \leq \theta \leq 360^\circ$ 2
- e) Write $1 - \frac{a-b}{a+b}$ as a single fraction in simplest form 2

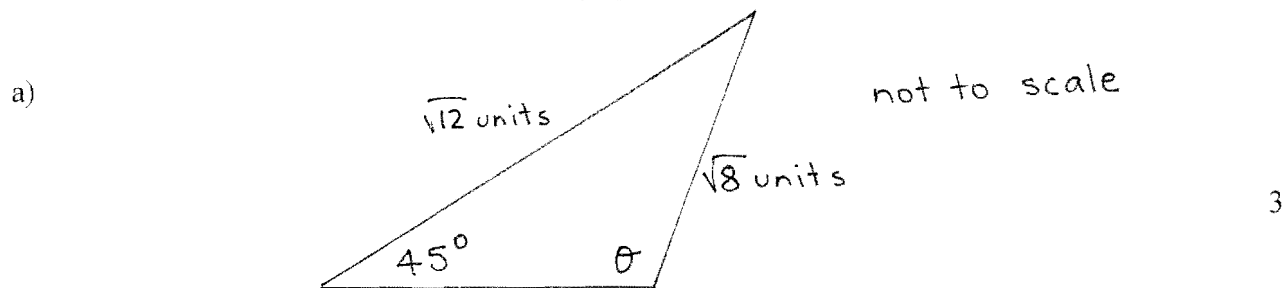
Question 2 (10 marks) Start a new page

- a) i) Sketch $y = 2 - (x + 1)^2$ clearly label the vertex and the y intercept 2
- ii) State the range for $y = 2 - (x + 1)^2$ 1
- b) The points A, B and C have coordinates (1,5), (6,0) and (5,7) respectively.
Plot these points on a number plane.
- i) Show that the length of AB is $5\sqrt{2}$ units. 1
- ii) Show that the triangle ABC is isosceles by finding the length of BC. 1
- iii) Find the equation of the line AB. 2
- iv) BA is produced to meet line $y = 7$ at P.
Find the coordinates of P. 1
- v) Find the area of triangle PAC. 2

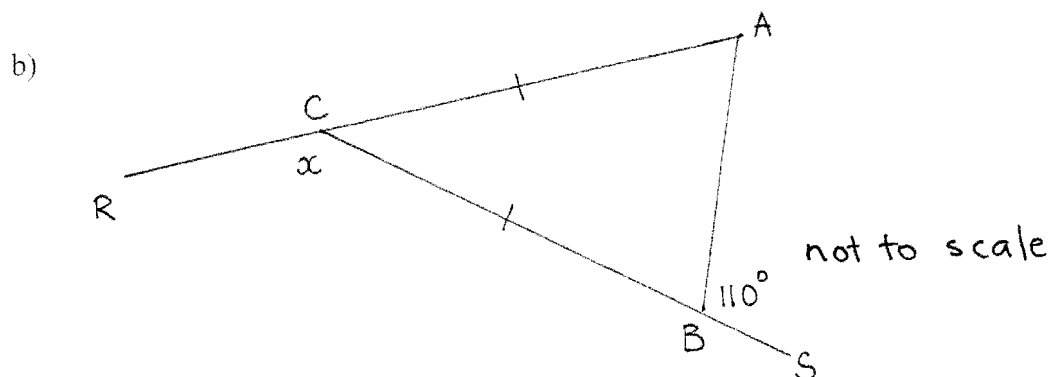
Question 3 (10 marks) (Start a new page)

- a) Express $0.3 + 0.\dot{3}$ in the form $\frac{a}{b}$ where a and b are integers 2
- b) Differentiate the following
- i) $y = \frac{2}{\sqrt{x}}$ 1
- ii) $y = \frac{3}{x-1}$ 2
- iii) $y = \frac{2x+1}{1-x}$ 2
- c) If α and β are the roots of the quadratic equation $x^2 - 5x + 2 = 0$ find the value of 3
- i) $\alpha + \beta$
- ii) $\alpha \beta$
- iii) $\alpha^2 + \beta^2$

Question 4 (10 marks) (Start a new page)



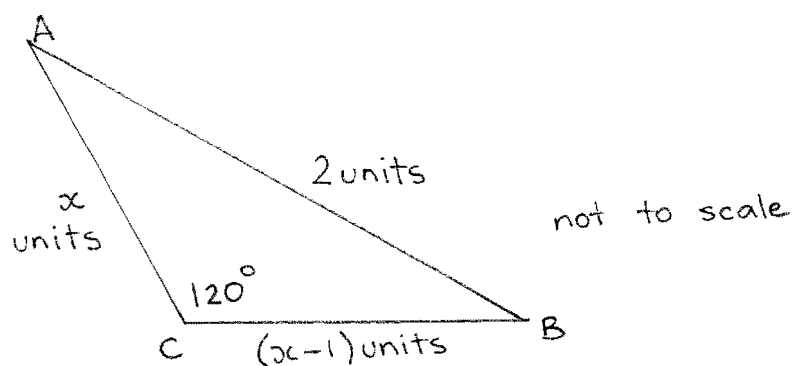
Use the sine rule to find the value of θ , where θ is obtuse



In the diagram above $AC = BC$, $\widehat{ABS} = 110^\circ$ and $\widehat{RCB} = x$.

Copy the diagram onto your answer sheets. Find the value of x giving reasons.

c)



In the diagram above, ABC is a triangle in which $AC = x$ units, $BC = (x - 1)$ units and $AB = 2$ units, $\angle ACB = 120^\circ$.

Find the exact value of x

3

d) Solve the pair of simultaneous equations

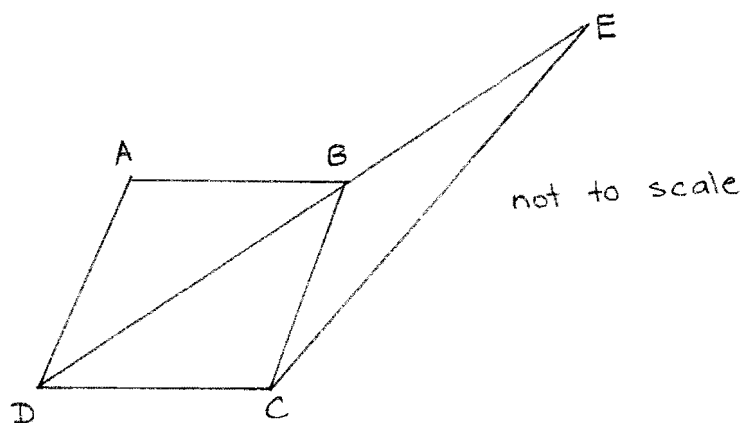
$$3x - y = 15$$

$$x + 2y = -2$$

2

Question 5 (10 marks) (Start a new page)

a)



In the diagram above, $ABCD$ is a rhombus.

DB is produced to E such that $\angle DEC = 10^\circ$, $\angle BDC = 30^\circ$

Copy the diagram showing the above information.

Calculate the size of $\angle BCE$ giving reasons

3

- b) Solve $2\cos^2 x - 3\cos x - 2 = 0$ for $-180^\circ \leq x \leq 180^\circ$ 3
- c) Prove $\frac{1}{\cot^2 \theta} + 1 = \sec^2 \theta$ 2
- d) Prove that $5x - 12y + 52 = 0$ is a tangent to the circle with centre the origin and radius 4 2

Question 6 (10 marks) (Start a new page)

- a) i) On the same diagram sketch $y = x - 4$ and $xy = 5$ (use a ruler for your axes) Label each function clearly 2
- ii) Find the points of intersection of these graphs 2
- b) A piecemeal function is defined as $f(x) = \begin{cases} 1 - x, & x < 0 \\ x^2 + 2, & x \geq 0 \end{cases}$
- i) Sketch the function 2
- ii) Find the value of $f(-2) + f(0) + f(3)$ 1
- iii) For what value(s) of x is this function not differentiable 1
- c) Find the value of k if the equation $x^2 - 6x + 4k = 0$ has real roots 2

Question 7 (10 marks) (Start a new page)

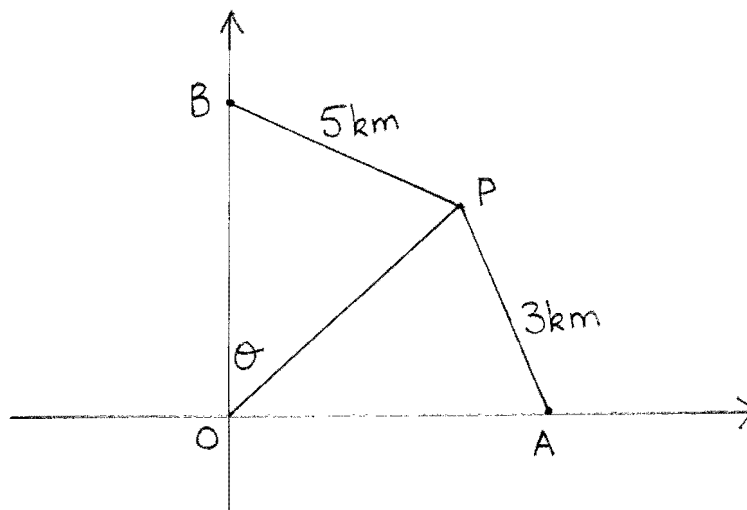
- a) Differentiate $y = x^2(1 - 2x)^3$ 2
- b) Find $\lim_{x \rightarrow 2} \left(\frac{x^2 - 4}{x - 2} \right)$ 1
- c) If $f(x) = x^2 - 3x + 5$, for what value of x is $f'(x) = 9$ 2
- d) The curve $y = ax + \frac{b}{x^2}$ cuts the x axis at $x = 2$, and the gradient of the tangent to the curve at this point is 3. Find the values of a and b . 3
- e) By first substituting m for x^2 , solve the equation $x^4 - 10x^2 + 9 = 0$ for x 2

Question 8 (10 Marks) (Start a new page)

a) For what values of a will $ax^2 + 5x + a$ be positive definite

2

b)



In the diagram above, A is a point due east of O and B is a point due north of O . P is on a bearing of 120° from B and 345° from A . AP is 3 km and BP is 5 km. The bearing of P from O is θ .

i) Find the size of $\hat{OBP}$ and $\hat{OAP}$

2

ii) Prove that $OP = \frac{5\sqrt{3}}{2\sin\theta}$

2

iii) Prove that $OP = \frac{3\sin 75^\circ}{\cos\theta}$

2

iv) Using parts ii) and iii) find θ to the nearest degree

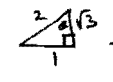
2

Question 1

a) $3^{2.5} = 15.6$ (3 sig fig)

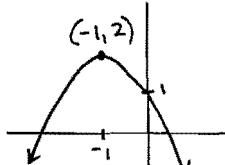
b) $ab - a - b + a$
 $= a(b-1) - a(b-1)$
 $= (b-1)(a-a)$

c) $\frac{1}{\sqrt{3}+2} \times \frac{\sqrt{3}-2}{\sqrt{3}-2} = \frac{\sqrt{3}-2}{3-4}$
 $= -\sqrt{3}+2$
 $\therefore a = -1 \quad b = 2$

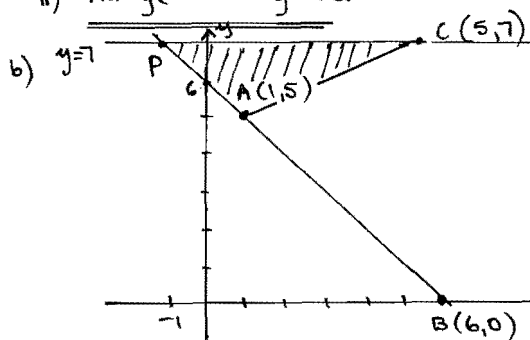
d) $\tan \theta = -\frac{1}{\sqrt{3}}$ 
 $\therefore \theta = 150^\circ, 330^\circ$

e) $1 - \frac{a-b}{a+b} = \frac{(a+b)-(a-b)}{a+b}$
 $= \frac{2b}{a+b}$

Question 2

a) i)  Vertex $(-1, 2)$
y intercept $y = 1$

ii) Range $y \leq 2$



i) $AB = \sqrt{(6-1)^2 + (0-5)^2}$
 $= \sqrt{25+25}$
 $= \sqrt{50}$
 $= 5\sqrt{2}$ units

ii) $BC = \sqrt{(6-5)^2 + (0-7)^2}$
 $= \sqrt{1+49}$
 $= \sqrt{50}$
 $= 5\sqrt{2}$

$AB = BC \therefore \triangle ABC$ is isosceles

iii) $m_{AB} = \frac{5-0}{1-6} = \frac{5}{-5} = -1$
eqn AB: $y-5 = -1(x-1)$
 $y = -x+6$

iv) $P(-1, 7)$

v) Area $\triangle PAC = \frac{6 \times 2}{2}$
 $= 6$ units²

Question 3

a) $0.3 + 0.3$
 $\frac{3}{10} + \frac{1}{3} = \frac{9+10}{30}$
 $= \frac{19}{30}$

b) i) $y = \frac{2}{\sqrt{x}} = 2x^{-1/2}$
 $\frac{dy}{dx} = 2x^{-1/2} \times -\frac{1}{2} x^{-3/2}$
 $= -\frac{1}{\sqrt{x^3}}$

ii) $y = 3(x-1)^{-1}$
 $\frac{dy}{dx} = 3x^{-1} (x-1)^{-2}$
 $= \frac{-3}{(x-1)^2}$

iii) $u = 2x+1 \quad v = 1-x$
 $u' = 2 \quad v' = -1$
 $\frac{dy}{dx} = \frac{2(1-x) - 1(2x+1)}{(1-x)^2}$
 $= \frac{2-2x+2x+1}{(1-x)^2}$
 $= \frac{3}{(1-x)^2}$

c) i) $\alpha + \beta = -\frac{b}{a} = 5$

ii) $\alpha \beta = \frac{c}{a} = 2$

iii) $\alpha^2 + \beta^2 = (\alpha + \beta)^2 - 2\alpha\beta$
 $= 25 - 4$
 $= 21$

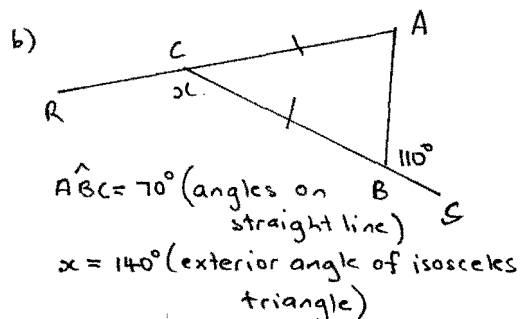
Question 4

a) $\frac{\sin \theta}{\sqrt{2}} = \frac{\sin 45^\circ}{\sqrt{8}}$
 $\sin \theta = \sqrt{2} \cdot \frac{1}{\sqrt{2}} \cdot \frac{1}{\sqrt{8}}$

$\sin \theta = \frac{2\sqrt{3}}{\sqrt{2} \cdot 2\sqrt{2}}$
 $\sin \theta = \frac{\sqrt{3}}{2}$

$\therefore$ acute $\theta = 60^\circ$

obtuse $\theta = 120^\circ$



c) $2^2 = x^2 + (x-1)^2 - 2x(x-1) \cos 120^\circ$
 $4 = x^2 + x^2 - 2x + 1 - 2x(x-1)(-\frac{1}{2})$
 $4 = 2x^2 - 2x + 1 + x^2 - x$
 $0 = 3x^2 - 3x - 3$
 $0 = x^2 - x - 1$
 $x = \frac{1 \pm \sqrt{1-4 \times 1 \times -1}}{2}$

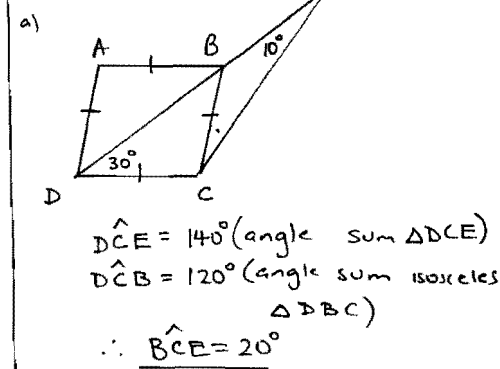
$x = \frac{1 \pm \sqrt{5}}{2}$ since $x > 0$

$\therefore x = \frac{1+\sqrt{5}}{2}$ units only

d) $3x - y = 15$ — ①
 $x + 2y = -2$ — ②

① $\times 2 \quad 6x - 2y = 30$
 $x + 2y = -2$ (+)
 $7x = 28$
 $\therefore x = 4 \quad y = -3$

Question 5



b) $2\cos^2 x - 3\cos x - 2 = 0$
 $(2\cos x + 1)(\cos x - 2) = 0$
 $\cos x = -\frac{1}{2} \quad \cos x = 2$ no sol.
 $\therefore$ acute $x = 60^\circ$
 $\therefore x = 120^\circ, -120^\circ$